USC Dornsife Spatial Sciences Institute
Master's Thesis Style Guide
Effective for Students in SSCI 594a as of Spring 2023

Turabian's (2018) Ninth Edition of *A Manual for Writers of Research Papers, Theses, and Dissertations* is used as the standard reference for style and format for the SSI M.S. programs. Turabian states general rules; We clarify herein the SSI requirements within those general rules and the occasional exception to them. Some exceptions are specific to the SSI and others are made to conform to the USC Graduate School's Thesis and Dissertation Guidelines (the "USC Guidelines"). Turabian is based on the Chicago Manual of Style, and you may refer to it for guidance on something not covered here or in Turabian.

Use this guide along with the SSI's GIST Thesis Template document (the "Thesis Template") to draft your thesis. The Thesis Template contains a number of "styles" stored in Microsoft Word that will help you fulfill many of the formatting requirements stated herein easily and consistently. Create your thesis document directly from the Thesis Template, using a "save as" in Microsoft Word. Read this document before and refer back to it as you write to conform to the USC Guidelines and the SSI requirements.

Section 1 of this guide contains general formatting requirements while section 2 specifies rules for individual elements. We refer to corresponding sections of Turabian throughout.

WHOLE-DOCUMENT FORMATTING REQUIREMENTS

This section contains formatting requirements that apply throughout the thesis document. We begin with citation formatting, since it is important to academic writing and may be new to many students, and then cover general non-citation formatting requirements that apply throughout the document.

1.1 Citation of Sources

Chapters 18 and 19 of Turabian (2018) describe in detail the style for citation formatting required of SSI theses. You must refer to your sources throughout your document using the "author-date" style, which is composed of two parts: abbreviated citations throughout the text of your document (called "parenthetical" citation) and full citation at the end of your document in the References section. You should get to know these two chapters very well.

Many students begin their thesis drafting by referring to related literature in the following type of way:

> In 2010, Matthew Zook, Mark Graham, Taylor Shelton, and Sean Gorman wrote "Volunteered Geographic Information And Crowdsourcing Disaster Relief: A Case Study Of The Haitian Earthquake," which describes the role of crowdsourced geospatial data in the recovery efforts in Haiti.

You should almost never provide the full title of a work in the text of your thesis. Similarly, you should almost never provide first names of authors in the text of your thesis. The Turabian style saves all of this information for the full references in the back of document and keeps the body of the text streamlined. A better textual reference to the above source is:

> Crowdsourced geospatial data played a key part in recovery efforts following Haiti's 2010 earthquake (Zook et al. 2010).

The sentence is set up so that it shares information about the topic of study and provides a cite to the source of that information. Compare this with the above, longer sentence, in which the sentence is structured so that the information shared is the fact that certain authors published

something. Note also the use of "et al." in the citation; Turabian requires et al. for parenthetical citations to sources with four or more authors.

1.2 General Formatting Requirements

This section details aspects of formatting that apply throughout the thesis document.

1.2.1 Margins (See Turabian A.1.1)

USC Graduate School guidelines require a one-inch margin on all four edges of the page. Thus, be sure that all tables and figures, in addition to your text, are within those margins. The only exception is for the page number (see 1.2.4).

1.2.2 Typeface (See Turabian A.1.2)

Use 12-point Times New Roman font throughout your document, including within tables and captions. You may make the font smaller in tables and maps, but the font should never be smaller than 9-point (USC Guidelines). This basic typeface requirement is stored in the relevant styles of the Thesis Template.

1.2.3 Spacing (See Turabian A.1.3)

Use only one space, not two, between sentences (i.e., when you finish writing a sentence, hit the space bar once, not twice).

Use double spacing for all lines of text except in the following circumstances:

- Use single spacing for:
  - Block quotations of **four** or more lines
  - Table and figure captions
  - Lists in appendices
- The contents of the following front and back matter elements should be single-spaced internally but with a blank line between items:
  - Table of Contents
  - Lists of Figures, Tables, or Abbreviations
  - List of References

1.2.4 Pagination (See Turabian A.1.4)

This section on pagination first explains what number to assign each page and then where these numbers should be placed on the page.

(i) Numbering

Theses have three main divisions: (1) front matter, (2) the body of the document, and (3) back matter. The front matter contains elements such as the Title Page and Table of Contents. The body is divided into chapters and, within each chapter, into sections and sub-sections. The back matter contains elements such as the References.

Number pages of the front matter with lowercase roman numerals (i, ii, iii, etc.) and the body of the document and the back matter with arabic numerals (1, 2, 3, etc.). Every page displays its page number except for the Title Page. Thus, the page following the Title Page (usually the Dedication) will display "ii" as its page number. The Thesis Template has this set-up already stored within it.

Number front matter separately from the rest of the text (i.e. restart numbering with the first page of Chapter 1). The Thesis Template includes a section break in Word to accomplish this task. The rest of the text, including back matter (References and Appendices), is numbered consecutively (i.e. do not restart numbering when you reach the back matter).

(ii) Placement

Page numbers are placed at the right margin in the footer (at the bottom of the page), a half inch from the bottom edge of the page.

For pages in landscape orientation, the page number must be in the same location and in the same direction as your portrait orientation pages. See the USC Guidelines and the Thesis Template for examples and instructions.

1.2.5 Titles of Main Elements (See Turabian A.1.5)

Titles of main elements must be formatted as followed (stored in Heading 1 style):

- 14-pt. Times New Roman font
- Bold
- Centered on the page

Place your titles at the top of the page, just below the margin.

The Heading 1 style includes the word "Chapter" followed by a number in an ordered outline. For chapter titles, keep this language and then type in the descriptive name of the chapter (i.e. "Related Literature").

For non-chapter titles, delete the word Chapter and the ordered list number and then type in your chapter title; this includes: Acknowledgments, Table of Contents, Lists of Tables, Figures, and Abbreviations, Abstract, List of References, and Appendices.

The only main elements which do not have a title are the optional Dedication and the Title Page. (The Title Page provides the title of the overall document, but does not contain a title for itself, i.e. "Title Page". For the formatting of the Title Page, see 2.1.1.)

1.2.6 Text

The text throughout your document is divided into paragraphs and formatted with the following specifications, which are stored in the Normal style of the Thesis Template:

- 12-pt. Times New Roman (see 1.2.2)
- Double-spacing (see 1.2.3)
- Half-inch, first-line indent
- No extra spacing before or after other paragraphs

The only exception to this textual formatting is for the first paragraph of each main textual element – Acknowledgements, Abstract, and each chapter. These first paragraphs have a first line that is flush with the left margin and thus no first-line indent. This is stored in the Normal No Indent style of the Thesis Template.

The text of an optional Dedication is slightly different and is discussed in 2.1.2.

2 FORMATTING REQUIREMENTS FOR SPECIFIC ELEMENTS

Individual elements of the thesis have their own formatting requirements. The elements are listed herein in the order which the USC Guidelines mandate they appear in your thesis document.

2.1 Front Matter

As noted, the front matter consists of the document elements which precede the body of the document. We list them here as they should appear in your document. Note that this order differs from how they are covered in the Turabian Appendix.

2.1.1 Title Page

See the Thesis Template and the USC Guidelines for formatting and spacing of elements on the Title Page. These contain the following formatting:

- Thesis title and optional subtitle are in 12-point font, headline-style capitalization (See Turabian 22.3.1 for a description of this style, as well as of sentence-style capitalization), centered, three inches from the top of the page (2" from top margin)
- The word "by" follows the title/subtitle by two single-spaced lines and is centered

- Your full name (no initials) follows the word “by” by two single-spaced lines and is centered
- The following block of text follows your name by seven single-spaced lines and is centered on the page as shown below:

A Thesis Presented to the
FACULTY OF THE USC DORNSIFE COLLEGE OF LETTERS, ARTS AND SCIENCES
UNIVERSITY OF SOUTHERN CALIFORNIA
In Partial Fulfillment of the
Requirements for the Degree
MASTER OF SCIENCE
(GEOGRAPHIC INFORMATION SCIENCE AND TECHNOLOGY)

- The month and year follow the block above by five single-spaced lines and are centered on the page. Degrees are only conferred in May, August, or December. Unless you complete early in any semester, your conferral month and year will the following semester: i.e. if you uploaded your thesis in the last half of the Fall 2022 semester, your conferral would be April 2023
- The line just above the bottom margin has copyright information flush left and your full name flush right. Copyright is in following format: Copyright 20XX

2.1.2 Epigraph. (See Turabian A.2.1.6)

An epigraph is short quotation or statement. It is optional in USC theses and not normally included in SSI GIST theses.

2.1.3 Dedication. (See Turabian A.2.1.4)

This element is optional but is often included in SSI GIST theses. Per the USC Guidelines, this element follows an optional Epigraph. Because an Epigraph is rarely included in a GIST thesis, the Dedication is thus usually page “ii”.

As noted above, the Dedication does not contain a title, but it does contain a page number, in roman numerals (see 1.6.1). The text is in 12-pt. Times New Roman font but otherwise differs from the Normal style in that it begins a third of the way down the page, is centered, and does not include a terminal period. The Thesis Template includes example language in the proper place on the page.

2.1.4 Acknowledgements. (See Turabian A.2.1.10)

GIST students often include an Acknowledgement in their thesis, to recognize individuals such as faculty, mentors, or fellow students, or to name institutions that supported their research in some way, such as by providing data or access to technology.

The title is in the Heading 1 style (see 1.7) and page numbers are in roman numerals (see 1.6.1).

Leave two double-spaced lines between the title and the first line of text. The first (and, often, only) paragraph is in the Normal No Indent style (see 1.2.6), and any following paragraphs are in the Normal style (see 1.2.6).

If the Acknowledgements stretch to more than one page, do not repeat the title.

2.1.5 Table of Contents. (See Turabian A.2.1.7)

The Table of Contents lists the title and beginning page number of all elements of your thesis document except two: the Title Page and the Table of Contents itself. This includes all

other elements of the front matter, each chapter and its sub-sections, and each element of the back matter.

The title is in the Heading 1 style (see 1.7) and page numbers are in roman numerals (see 1.6).

Leave two single-spaced lines between the title and the first item listed. Main items should have one double-spaced line between them, while all sub-headings of a chapter should be single-spaced (see 1.2.3). The Thesis Template contains an example Table of Contents with proper spacing.

Provide page numbers only for the first page of each element (not the full span of pages), and use lowercase roman or arabic numerals as on the pages themselves.

Main elements are flush with the left margin. Chapter sections and sub-sections are indented consecutively.

Use a line of periods or ellipses (called leaders) to lead a reader's eye from each title to the page number. If you use Microsoft Word to auto-generate these lists based on your stored captions and table titles, these will be included.

If the Table of Contents stretches to more than one page, do not repeat the title.

The Thesis Template document includes proper formatting for auto-generated Table of Contents, so it is highly recommended that you use it. If you consistently apply the proper Heading styles throughout your document, you can easily populate and re-populate the Table of Contents to quickly reflect any changes in pagination.

2.1.6 List of Tables and Figures. (See Turabian A.2.1.8)

Your GIST thesis will undoubtedly contain tables and figures, and your thesis will therefore require both a List of Tables and a List of Figures in the front matter. As noted above, each of these elements will have a title in the Heading 1 style (see 1.7) and page numbers in roman numerals (see 1.6).

Include two double-spaced lines between the title and the first item listed. Single-space individual items listed but leave a blank line between items (see 1.2.3)

Figure captions and table titles should match the wording and capitalization of each in the thesis document itself, so try to keep them short enough in the paper to avoid multiple lines in the list. However, if they are very long, you may shorten them in a logical way in this list. (See Turabian 26.2.2 and 26.3.2 for more on table titles and figure captions.)

List page numbers flush right and use leaders (see A.2.1.7) to connect the captions and titles to page numbers. If you use Microsoft Word to auto-generate these lists based on your stored captions and table titles, these will be included.

If either list is more than one page, do not repeat the title.

2.1.7 List of Abbreviations. (See Turabian A.2.1.11)

If your thesis employs more than a few abbreviations, provide a List of Abbreviations in the front matter. If you include a List of Abbreviations, it must list all abbreviations that you refer to in the text.

The title is in the Heading 1 style (see 1.7) and page numbers are in roman numerals (see 1.6).

Include two double-spaced lines between the title and the first item listed. Single-space individual items listed but leave a blank line between items (see 1.2.3).

List items alphabetically by the abbreviation, not by the spelled-out term. The abbreviations themselves are flush left; spelled-out terms are set on a consistent indent that

allows about a half inch of space between the longest abbreviation in the first column and the first word in the second column.

If the list stretches to more than one page, do not repeat the title.

2.1.8 Abstract. (See Turabian A.2.1.4)

An Abstract summarizes the thesis contents, including a summary of the results.

The title is in the Heading 1 style (see 1.7) and page numbers are in roman numerals (see 1.6). Include two double-spaced lines between the title and the first line of text.

Most Abstracts are composed of one paragraph of text. This paragraph is in the Normal No Indent style (see 1.2.6). You may break your abstract into two or more paragraphs if it improves readability. The second paragraph would be in the Normal style (see 1.2.6).

Do not include citations in your abstract.

Only define abbreviations in the Abstract if you refer to them again in the Abstract.

The Abstract should generally be no more than one page, but if it does stretch to a second page, do not repeat the title.

2.2 Body

The body of the thesis includes everything between the front matter and the back matter. The main text is separated into chapters and, further, into sections and subsections. Figures and tables will be inserted throughout. Specific requirements for main text, chapters, sections, figures and tables follow.

2.2.1 Main Text

The main text will generally be divided into paragraphs, and there are few additional format requirements beyond what has been described above (see e.g. the discussion of the Normal and Normal No Indent styles in 1.2.6).

As a general rule, footnotes and endnotes are to be avoided. Incorporate explanatory information succinctly in the main text, and leave out non-relevant information.

If you abbreviate the title of something (an organization, a statute, etc.), define the abbreviation the first time you refer to the title and then consistently refer to the abbreviation rather than full title throughout the rest of the document.

Only include URLs in the text if they are quite short. Follow the rules laid out in Turabian Ch. 19 for citation to online sources. Suppress live links for URLs.

Create equations and formulas with the Microsoft Word Equation Editor if possible. If not, use a different program and insert equations into your file as images. Leave one blank line between the equation and the surrounding text. Center the equation on the page. If you have several equations, or you refer to them individually later in the text, number each equation sequentially with the number placed in parentheses, e.g. (3), flush to the right margin. Discussion about numbered equations should be stated as, for example: “Equation 3.” If you have only one or two equations, no number is required.

2.2.2 Chapters. (see Turabian A.2.2.3)

GIST theses normally consist of five chapters: Introduction, Related Literature, Methods, Results, and Conclusions. Web GIS projects often break up the single Methods chapter into two with one focusing on topics such as user requirements, choice of software, and data, and one focusing on the development process.

Each chapter begins on a new page. Chapter titles are in the Heading 1 style (see 1.7). They begin with “Chapter #” with the number as an arabic numeral, followed by a space and then the chapter’s descriptive title, in headline-style capitalization (See Turabian 22.3.1 for a description of this style).

Leave two blank lines between the title and the first line of text.

Always include at least one paragraph of text between the chapter title and the first section heading. This text should introduce the contents of the chapter, not reiterate an introduction to the project overall or anything else you have already stated.

Do not repeat the number or the title on subsequent pages of the chapter.

2.2.3 Sections and Subsections. (see Turabian A.2.2.4)

Chapters are divided into sections, which in tum may be divided into subsections.

The SSI required style for three levels of sections/subsections follow:

- First level:
  - 14-point font; bold; flush left; 12-point spacing above and below; Outline numbering at level 2 (stored in the Thesis Template as Heading 2)
  - Use headline-style capitalization, with no terminal period
- Second level:
  - 12-point font; italics; flush left; 12-point spacing above and below; Outline numbering at level 3 (stored in the Thesis Template as Heading 3)
  - Use headline-style capitalization, with no terminal period
- Third level:
  - 12-point font; flush left; 12-point spacing above and below; Outline numbering at level 4 (stored in the Thesis Template as Heading 4)
  - Use sentence-style capitalization, with no terminal period

Do not use more than three levels of sectioning. Rather than being helpful, it becomes distracting.

You should have at least two sections per chapter and at least two subsections within any section that has a subsection. Thus, if section 3 has a subsection 3.1, it must also have a subsection 3.2.

Just as you need introductory text between the chapter title and the first section heading which introduces the chapter, you must include some introductory text between a section heading and its first subsection that explains the scope of the section.

Keep section headings succinct.

Keep all headings attached to the ensuing paragraph. All the heading styles in the Thesis Template include the “keep with next” paragraph setting.

As noted, if you use Heading styles consistently and correctly, the Table of Contents with page numbers, can be automatically generated and updated as needed.

2.2.4 Tables. (see Turabian 26.2)

Tables are *de rigueur* in GIST theses. You will have at least a data table and, likely, a few others tables. The first section herein describes table placement and sizing, the second explains formatting of titles and sourcing information, and the third discusses the contents of a table.

(i) Placement and sizing of tables

Create tables with the table editor in Word or in a spreadsheet program such as Excel and insert them into your file as embedded tables. Do not insert tables as graphic images. You must retype them to make them consistent with the styles described herein.

Center tables on the page. Check the page ruler to be sure the centering is measured from the margins, rather than from any indent that might inadvertently be in place. The Table Centered style in the Thesis Template can help you achieve proper placement.

Tables should only be as wide as is necessary to share the content effectively. If a table needs to be wider than 7.5 inches, you may use landscape orientation (see 1.2.4(ii)).

Tables must be placed between paragraphs and not in the midst of a paragraph. A table should generally be placed directly following the paragraph in which it is first referred to. See Turabian 26.1.1 for direction on placing tables when such placement would create more than a few lines of blank space on a page.

Leave at least one blank line between the table title and any text above it on the page, and also between the bottom of the table (or source information) and any text below it.

(ii) Table titles and source information

All tables must have a numbered title: "Table [#]." followed by a succinct title in sentence-style capitalization, with no terminal period. The font of titles should match the body text: 12-point regular Times New Roman font (no bold or italics).

See the Thesis Template for instructions on creating table titles in Microsoft Word with the "caption" functionality so that you can auto-populate the List of Tables in the front matter.

The title should sit flush left with the edge of the table and run the length of the table. If the table is quite narrow, move the title margins beyond the edges of the table an equal amount on either side, up to one inch on either side.

Place source information, if any under the bottom line of a table, single spaced, flush with the left edge of the table.

Between the table and its title (and between the table and its source information), leave a 6-point space.

(iii)Table contents

Use 12-point font within tables. If necessary, you may use a smaller font, but it must be Times New Roman, and it must never be smaller than 9-point (see USC Guidelines).

Use .5 pt black lines for defining the rows, columns, and borders of your tables.

Do not use background shading within tables.

Consistency in table formatting throughout your document is important. The Thesis Template has two styles – Table Centered and Table Justified – to help keep the formatting of table contents consistent.

2.2.5 Figures. (see Turabian 26.3)

GIST theses will always contain figures, because maps can quickly and powerfully communicate spatial findings. Images can add depth to the description of a study area, while charts can effectively reveal various aspects of data. Figures, thus, do not replace text and are not filler for increasing one's page count while decreasing the burden of writing. Only include a figure if it truly adds to the meaning and effectiveness of your document.

All figures must be discussed in the text – not just pointed to. If you do not have at least a few sentences of content to say about a figure, then it does not belong in your document.

The first section discusses formatting rules that apply to all types of figures, and the following section describes specific requirements for maps.

(i) General figure formatting

Figures should be only as large as is necessary to clearly share their information. All text within figures, including maps and images, must be legible. Be careful when you compress graphics to fit on the page to ensure that font does not become smaller than 9-point or illegible.

Figures must be centered on the page and stay within the margins of the page. If a figure must be wider than 7.5 inches, you may use landscape orientation (see 1.2.4(ii)).

Give all figures a .75 pt black border.

All figures require a numbered caption: “Figure [#].” followed by a space, then a succinct description of the figure. Use sentence-style capitalization, with no terminal period, unless an additional sentence follows. The font should match the body text – 12-point regular Times New Roman font (no bold or italics) – and lines should be single-spaced (see 1.2.3).

Place a caption below its figure, flush left with the edge of the figure. Run the caption the full width of the figure. If the figure is quite narrow, move the margins beyond the edges of the figure an equal amount on either side, up to one inch on either side. You may also use one caption to refer to two or more small figures that you group together and refer to specifically within the caption using lower-case letters.

The Caption style in the Thesis Template contains these specifications, but you will need to manually adjust the margins of the caption. See the Thesis Template for instructions on creating captions in Microsoft Word with the “caption” functionality, so that you can auto-populate the List of Figures in the front matter.

Figures must be placed between paragraphs, not in the midst of a paragraph. A figure should generally be placed directly following the paragraph in which it is first referred to. See Turabian 26.1.1 for direction on placing figures when such placement would create more than a few lines of blank space on a page.

Leave at least one blank line (and preferably two) between the figure and any text above it on the page, and also between the caption and any text below it.

Between the figure and its caption, leave a 6-point space. The Figure style in the Thesis Template can help you achieve the placement and spacing for a figure.

(ii) Maps

Maps are important part of the GIST thesis and must be created with care.

Maps must contain a scale bar. The scale bar should only be as large as necessary to be effective and legible. Distance values should be round numbers (e.g. a scale bar with intervals of 10, 20, and 40 is appropriate but not one with 8, 16, and 32). Be very careful with shrinking or expanding maps once you have placed them in your document. If you must make the map figure bigger or smaller, be sure you have locked the aspect ratio, so that the scale bar continues to hold true across the map.

Maps should generally have a north arrow. It should be sized only as large as necessary to be clear.

Thematic maps will generally require a legend. Legend items must be listed with names that make sense – do not simply use the layer names of the data as your worked with them in the GIS. Formatting of legend items, including font, spacing between items, spacing within the border, and sizing of patches, must be clean and clear, and must be employed consistently throughout all maps in the document.

Maps do not require a title when they are a part of your thesis. The figure caption plays the role of the map title.

When necessary, maps should have inset context maps to help the reader know where the site is located. Similarly, use inset detail maps as necessary to better visualize key areas.

2.3 Back Matter (see Turabian A.2.3)

The back matter of your thesis will begin with your References and may contain one or more appendices.

2.3.1 References

Since you are using author-date citations, you must include a reference list in the back matter. Give the list a title of “References” in the Heading 1 style (see 1.7).

Include two double-spaced lines between the title and the first item listed. Single-space individual items listed but leave a blank line between items (see 1.2.3)

Individual items are listed flush with the left margin and employ a hanging indent of .5 inch.

Chapters 18 and 19 of Turabian (2018) provide extended detail on formatting citations and the reference list. There are some minor differences between Chicago and Turabian styles, so use Turabian's rules when there is a difference. Turabian rules must be strictly adhered to with one key exception: **In the reference list, provide first and middle initials – not names**. This exception is to match the citation formatting of leading journals in our field.

2.3.2 Appendices.

If your thesis includes essential supporting material that cannot be easily worked into the body of your paper, share the material in one or more appendices. Examples of such material are extra tables or maps of analytical results, a survey completed by research subjects or testers of a mobile mapping application, or instructions shared with volunteer mappers.

Make sure to include a brief description in the main body about the contents of every appendix, referring to them by number in a manner similar to figures and tables.

Appendix titles are listed at the top of the page in the Heading 1 style. If the appendix is more than one page, do not repeat the title. Leave two blank lines between the title and the first line of text or other material.

If the appendix material is of different types—for example, a table and a case study—divide it among more than one appendix.

When there are multiple appendices, give each appendix a letter and a descriptive title. Use single, capitalized letters of the alphabet in sequential order (A, B). Put the number or letter following the word Appendix, follow with a space and the descriptive title in headline-style capitalization. (If your paper has only one appendix, you may also give it a descriptive title, but do not give it a number or letter.)

Do not number tables or figures in appendices; Refer to them collectively by the appendix number only.